

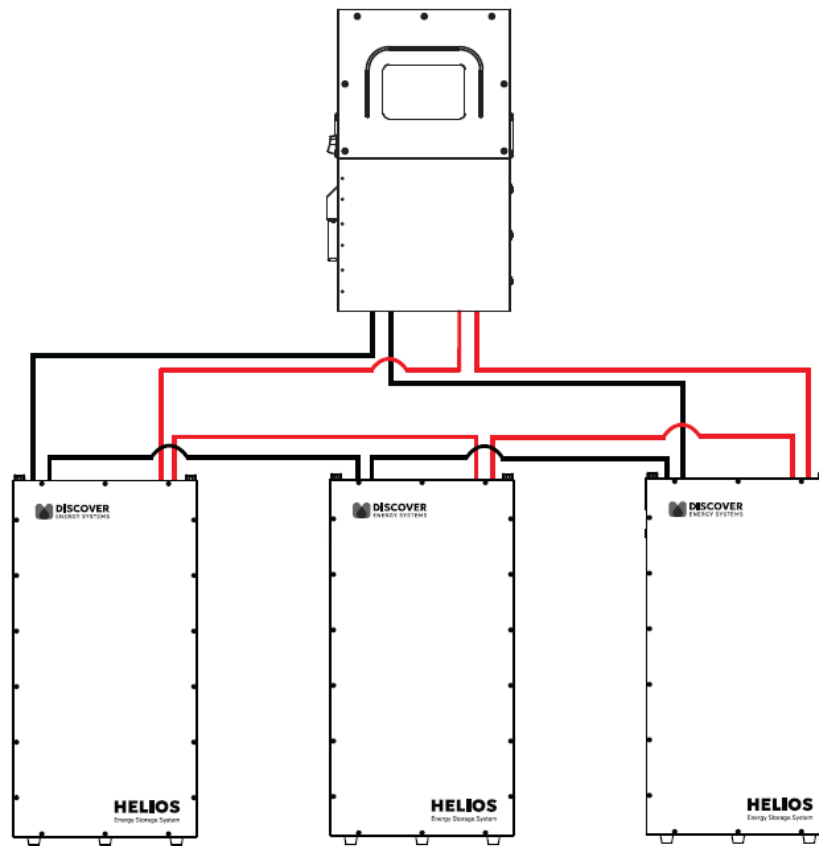


HELIOS ESS Parallel Installation

Revision REV B

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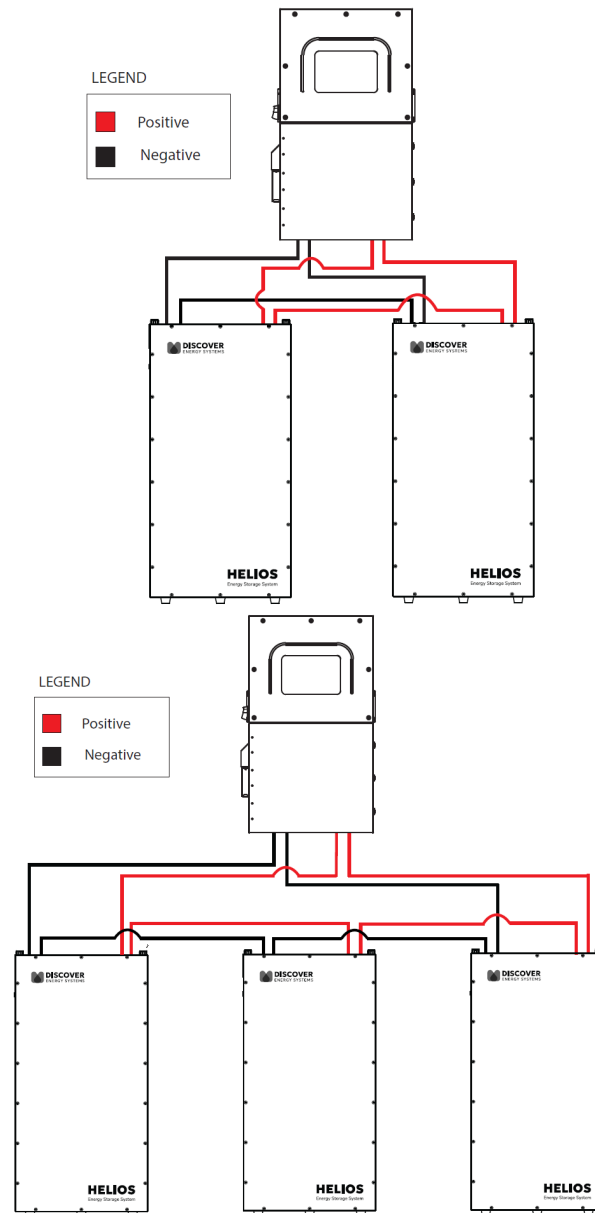
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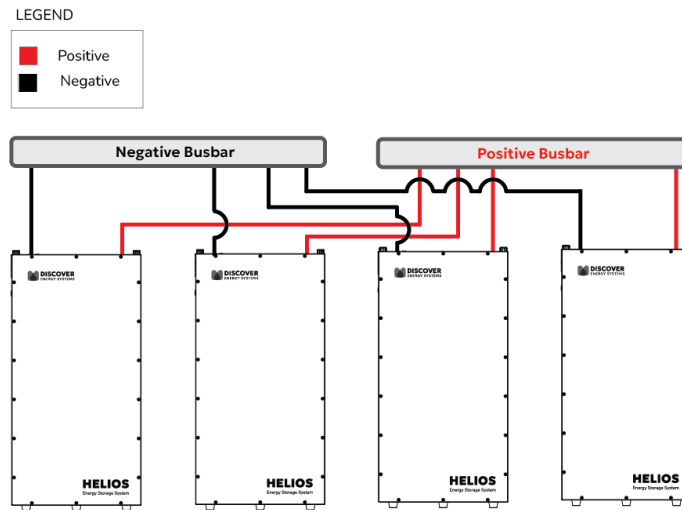


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1 INTRODUCTION

This document describes how to connect the HELIOS ESS, a rechargeable Li-ion battery from Discover Energy Systems (Discover), in a battery bank configuration where multiple batteries are connected in parallel.





The following accessories are available for connecting the HELIOS ESS batteries in parallel.

Accessory	Part Number
HELIOS ESS Battery to Inverter Power Cables 2 pairs of 200 A cables with quick connects: 2/0 AWG Positive Battery Cable (1.0 m, 39.37 in) 2/0 AWG Negative Battery Cable (1.0 m, 39.37 in) 2 pairs of 170 A cables with quick connects: 1/0 AWG Positive Battery Cable (1.0 m, 39.37 in) 1/0 AWG Negative Battery Cable (1.0 m, 39.37 in)	950-0070 950-0071
HELIOS ESS Parallel Wire Kit (battery to battery) 1 pair of 200 A cables with quick connects: 2/0 AWG Positive Battery Cable (1.35 m, 53 in) 2/0 AWG Negative Battery Cable (1.35 m, 53 in) - CAT6 Cable (135 cm, 53 inch) (COM3/COM4)	950-0068
HELIOS ESS Terminal Connector Set	950-0072

2 AUDIENCE, SAFETY, MESSAGES, AND WARNINGS

2.1 Audience

Configuration, installations, service, and operating tasks should only be performed by qualified personnel in consultation with local authorities having jurisdiction and authorized dealers. Qualified personnel should have training, knowledge, and experience in the:

- Installation and commissioning of electrical equipment.
- Application of electrical codes, safety, and installation standards.
- Analysis and reduction of hazards involved in performing electrical work.
- Installation, configuration, and maintenance of batteries and energy storage systems.
- Installation and configuration of communication systems, such as CAN and Ethernet.
- Installation and configuration of systems activated by relays.

2.2 Warning, Caution, Notice, and Note Messages

Messages in this manual are formatted according to this structure.

	Additional information concerning important procedures and features of the product. Read all the instructions before installation, operation, and maintenance.
	Important information regarding hazardous conditions.

WARNING

Important information regarding hazardous conditions that may result in personal injury or death.

CAUTION

Important information regarding hazardous conditions that may result in personal injury.

NOTICE

Important information regarding conditions that may result in damage to the equipment but not personal injury.

NOTE

Ad hoc information concerning important procedures and features of the battery not related to personal injury or equipment damage.

2.3 General Warnings

 WARNING
ELECTRIC SHOCK AND FIRE HAZARD

- This equipment must only be installed as specified.
- Do not disassemble or modify the battery.
- If the battery case has been damaged, do not touch exposed contents.
- There are no user-serviceable parts inside.

Failure to follow these instructions may result in death or serious injury.

 WARNING
ELECTRIC SHOCK AND FIRE HAZARD

Do not lay tools or other metal parts on top of the battery or across the terminals.

Failure to follow these instructions may result in death or serious injury.

**CAUTION****ELECTRIC SHOCK**

- Do not touch the energized surfaces of any electrical component in the battery system.
- Before servicing the battery, follow all procedures to fully de-energize the battery system.
- Follow the “Safe Handling Procedures” below when working with the battery.

Failure to follow these instructions may result in injury.

2.4 Personal Protective Equipment

When handling or working near a battery:

- Use Personal Protective Equipment, including clothing, glasses, insulated gloves, and boots.
- Do not wear rings, watches, bracelets, or necklaces.

2.5 Safe Handling Procedures

Before using the battery and any power electronics, read all instructions and cautionary markings on all components and appropriate sections of their manuals.

- Use personal protective equipment when working with batteries.
- Do not dispose of the battery in fire.
- Promptly dispose or recycle used batteries following local regulations.
- Do not disassemble, open, crush, bend, deform, puncture or shred.
- Do not modify, re-manufacture, attempt to insert foreign objects into the battery, immerse or expose to water or other liquids, or expose to fire explosion or other hazards. If the user suspects damage to the battery due to water, heat, or other reason, take it to a service center for inspection.
- Only use the battery for the system for which it is specified.
- Do not lift or carry the battery while in operation.
- When lifting a heavy battery, follow the appropriate standards.
- Only lift, move, or mount following local regulations.
- Take care when handling battery terminals and cabling.
- Only use the battery with a charging system that has been qualified for the system. The use of an unqualified battery or charger may present a risk of fire, explosion, leakage, or other hazards.
- Do not short circuit a battery or allow metallic conductive objects to contact battery terminals.
- Replace the battery only with another battery that has been qualified for the system. Using an unqualified battery may present a risk of fire, explosion, leakage, or other hazards.

- Avoid dropping the device or battery. If the device or battery is dropped, especially on a hard surface, and the user suspects damage, take it to a service center for inspection.

3 HELIOS ESS - PARALLEL INSTALLATION

The instructions in this document describe how to connect multiple batteries in parallel to an inverter.

WARNING

ELECTRIC SHOCK AND FIRE HAZARD

- This equipment must only be installed as specified.
- Do not disassemble or modify the battery.
- Do not touch exposed contents if the battery case has been damaged.
- There are no user-serviceable parts inside.

Failure to follow these instructions may result in injury.

NOTICE

EQUIPMENT DAMAGE

- Some chargers and inverters support temperature-compensated charging. Disable temperature-compensated charging on the charger or inverter.
- Do not use or install a battery temperature sensor.

Failure to follow these instructions may result in equipment damage.

NOTE

It is the responsibility of the installer to ensure that all applicable installation requirements and standards are met.

3.1 Tools

- Insulated tools sized to match nuts, bolts, and cables
- True RMS Voltmeter
- Personal protective equipment

3.2 Wiring

3.2.1 DC Cables



CAUTION

FIRE HAZARD

Undersized cables can become hot and may potentially catch fire.

Failure to follow these instructions may result in injury.

Use the Battery to Inverter Power Cables 2/0 AWG (950-0070) or 1/0 AWG (950-0071) to wire the batteries to the inverter. Select the cables compatible with your inverter and inverter settings. You can also use your own DC cables by using the HELIOS ESS Connector Set (950-0072) if the cables are rated for the maximum voltage of any circuits sharing the same wiring space, sized according to local codes, and meet the following requirements.

1. **Copper-stranded cable.** DC cables must be stranded, copper, and rated 90 °C minimum. Terminate the cables on one side with lugs that fit the DC terminals on the inverter, and on the other side of the cable the plug and pull quick connects are used to mate with the plug and pull terminals on the HELIOS ESS battery.
2. **Minimum and equal cable lengths.** Select a location that minimizes the length of battery cables to reduce voltage drop from the impedance leading to reduced performance. When installing multiple batteries in parallel, the length of all the battery cables should be the same.
3. **Appropriate cable gauge.** The cables should be capable of carrying the normally expected current, plus a margin of safety.
4. **Proper polarity.** Positive (+) is connected to positive (+), and negative (-) is connected to negative (-). Verify the polarity of all connections before energizing batteries.

3.2.2 DC Protection

CAUTION

FIRE HAZARD

- Undersized fuses and disconnects may become overloaded and potentially cause a fire.
- Fuses and disconnects are required to open before the cable reaches its maximum current carrying capability.

Failure to follow these instructions may result in injury.

Each battery comes equipped with an integrated breaker that opens the positive pole. When multiple batteries are installed in parallel, additional DC overcurrent protection is recommended between the battery bank and inverter-charger.

1. **Appropriate size.** Size the fuses and disconnects following local codes to protect the wiring in the system. The fuses and disconnects are required to open before the cable reaches its maximum current carrying capability.
2. **Install protection in the positive cable.** The fuse and disconnect should be located close to the battery and installed in the positive cable. Local codes may limit how far the protection can be from the battery.

NOTE

- AC circuit breakers and DC circuit breakers are not interchangeable. Check the label on the circuit breaker to ensure it is the correct type and rating. Use only DC circuit breakers.
- The local electrical code in your area may require both positive and negative DC disconnect switches. Refer to your applicable code and check with the authority having jurisdiction to confirm local requirements.

3.2.3 Terminal Connections and Hardware

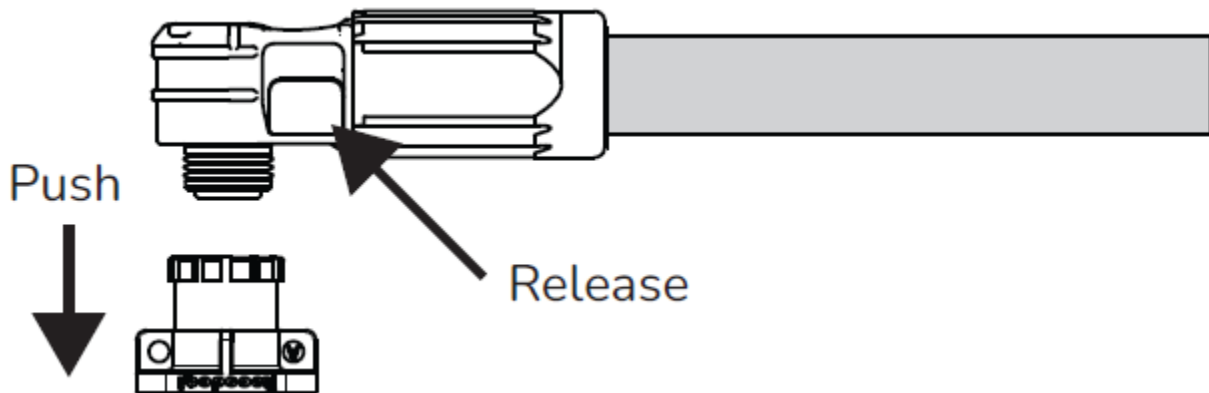
⚠ CAUTION

ELECTRIC SHOCK AND FIRE HAZARD

- Take precautions to avoid bridging the terminals.
- Do not contact the terminals with metal mountings, fixtures, or body parts.

Failure to follow these instructions may result in injury.

Plug-and-pull quick connects are used to mate with the plug-and-pull terminals on the HELIOS ESS battery.



1. Plug the connector into the plug-and-pull terminal.
2. Gently pull the connector to confirm it has snapped in place.
3. To disconnect, press the release button on the side of the connector.

3.3 Parallel Battery Installation

⚠ CAUTION

FIRE HAZARD

Use cables that meet specifications. Undersized cables can become hot and may potentially catch fire.

Failure to follow these instructions may result in injury.

Batteries used in a parallel system must meet the following requirements:

- **Same model.** Batteries must be the same model.
- **Equal voltage.** The battery has a Pre-Charge System that allows you to connect batteries at different SOC. The best practice is to confirm batteries are within 50 mV (0.05 V) of each other before installing them in parallel.

Cables interconnecting the batteries must meet the following requirements to optimize performance:

- **Minimum cable length.** Select a location that minimizes the length of battery cables to reduce voltage drop from the impedance.
- **Equal cable gauge.** The interconnection cables should be the same gauge.
- **Equal cable length.** The interconnection cables should be of equal length.
- **Connectors.** Battery terminal connections require specific connectors. Refer to the “HELIOS ESS Accessories For Parallel Installation” section.

Install equipment following the standards set by the local authority having jurisdiction. The following instructions assume the battery and inverter have been attached to the wall.

NOTICE

EQUIPMENT DAMAGE

- Avoid short circuits. Short circuits will damage the battery and void the warranty.
- Failure to securely lock the ends of battery cables to the receiving pins on the battery will increase resistance and lower voltage, leading to burnout of the terminals and voiding of the warranty.
- Some inverter-chargers support temperature-compensated charging. Disable temperature compensated charging on the charger or inverter.
- Do not use or install a battery temperature sensor.

Failure to follow these instructions may result in equipment damage.

NOTE

To connect HELIOS ESS batteries in parallel, use busbars, the HELIOS ESS Parallel Wire Kit (950-0068), or create your own appropriately-sized cables with the HELIOS ESS Terminal Connector Set (950-0072).

3.4 Parallel Battery Installation Procedure (Parallel Wiring Kit)

The following instructions identify how to connect batteries and inverters in parallel using the HELIOS ESS Parallel Wire Kit (950-0068). One kit is required to parallel two batteries. Two kits are required to parallel three batteries.

⚠ CAUTION**ELECTRIC SHOCK AND FIRE HAZARD**

- Take precautions to avoid bridging the terminals.
- Do not contact the terminals with metal mountings, fixtures, or body parts.

Failure to follow these instructions may result in injury.

NOTICE**UNBALANCED BATTERY CELLS**

Paralleling more than three batteries using the Parallel Wiring Kit may lead to unbalanced battery cells.

Failure to follow these instructions may result in equipment damage.

Install equipment following the standards set by the local authority having jurisdiction.

1. Prepare the inverter and battery for wiring.
 - a. If the inverter is wired to a power source, open the disconnect and set the inverter OFF.
 - b. Use a DMM or other voltage measuring device to confirm the circuit is de-energized.
 - c. If the circuit in which the battery is installed has a disconnect, open the disconnect to isolate the battery.
 - d. Set both the battery BMS and the breaker to the OFF position.
 - e. Use a DMM or other voltage measuring device to confirm the circuit is de-energized.

2. Wire the inverter.

The Battery to Inverter Power Cables 2/0 AWG (950-0070) or 1/0 AWG (950-0071) are available to wire the batteries to the inverter. Select the cable set compatible with your inverter and inverter settings.

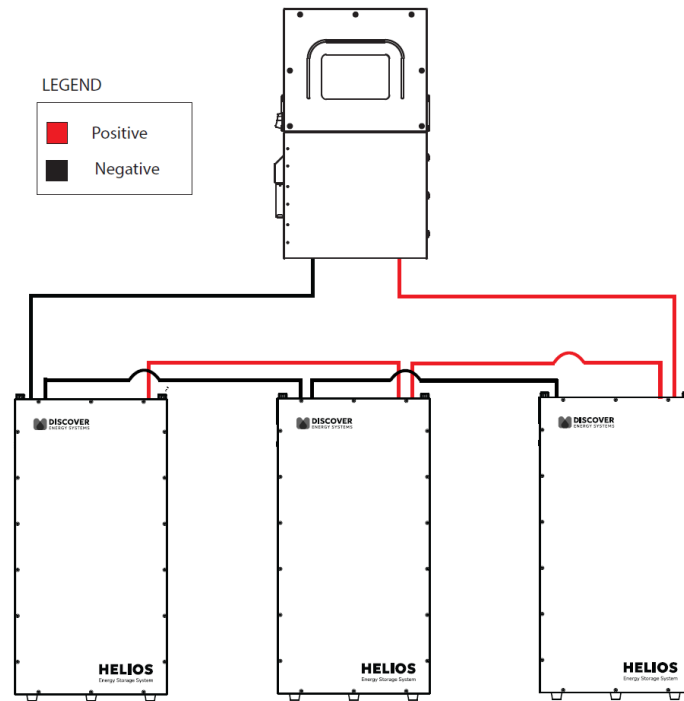
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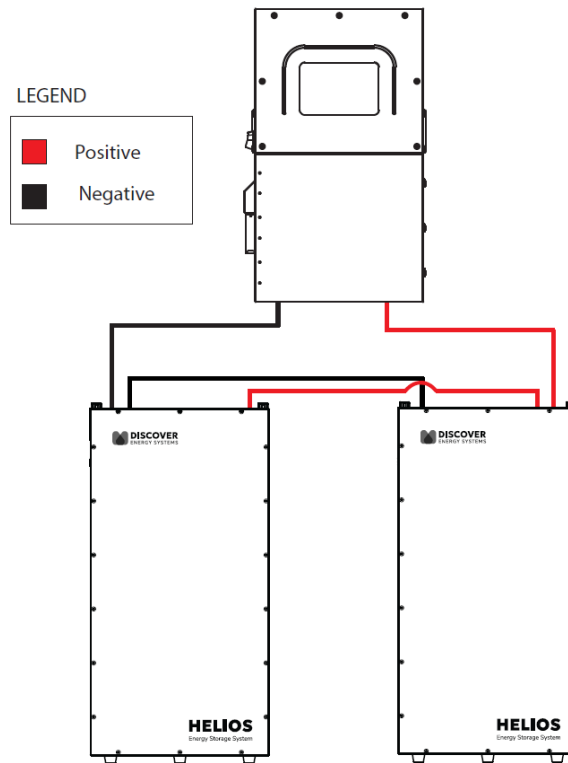
- When selecting the cables for use between the battery and the inverter-charger, note that a single pair of 2/0 AWG (950-0070) cables is suitable for current up to 200 A, and a single pair of 1/0 AWG (950-0071) cables is suitable for current up to 170 A.
- The selection between using two pairs or a single pair of cables between the inverter and the batteries, is based on safely matching the load of the charger (A) and the cable's maximum current carrying capacity.

3. Ensure the cable connections are clean and in working order.
 - a. Connect the positive battery cable to a positive battery terminal on the inverter.

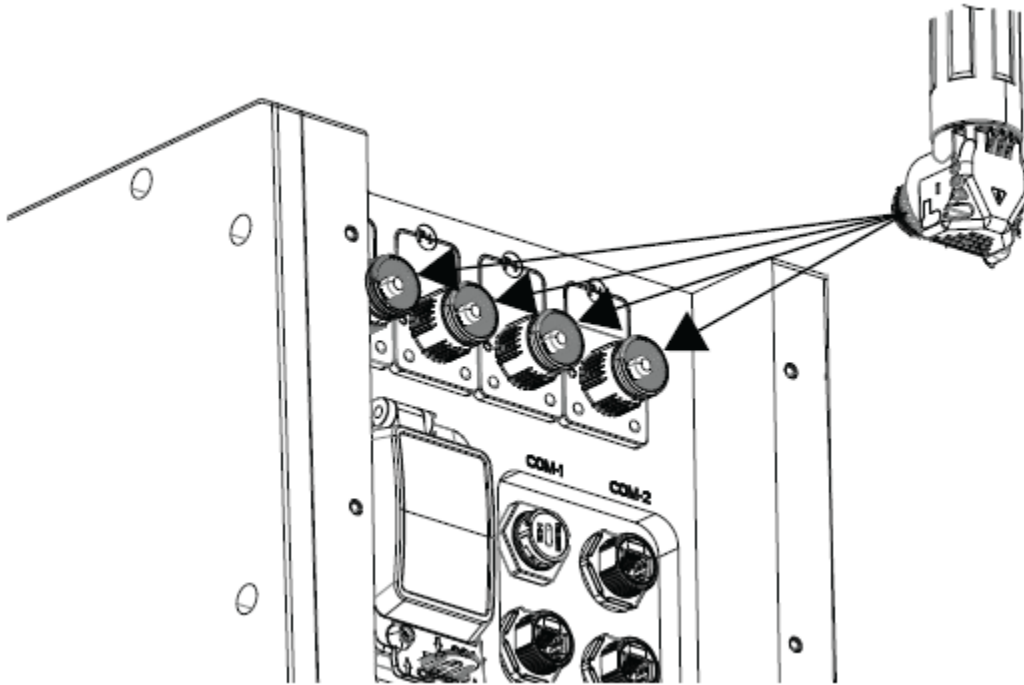
- b. Connect the negative battery cable to a negative terminal on the inverter.

When using ONE Set of Battery to Inverter cables



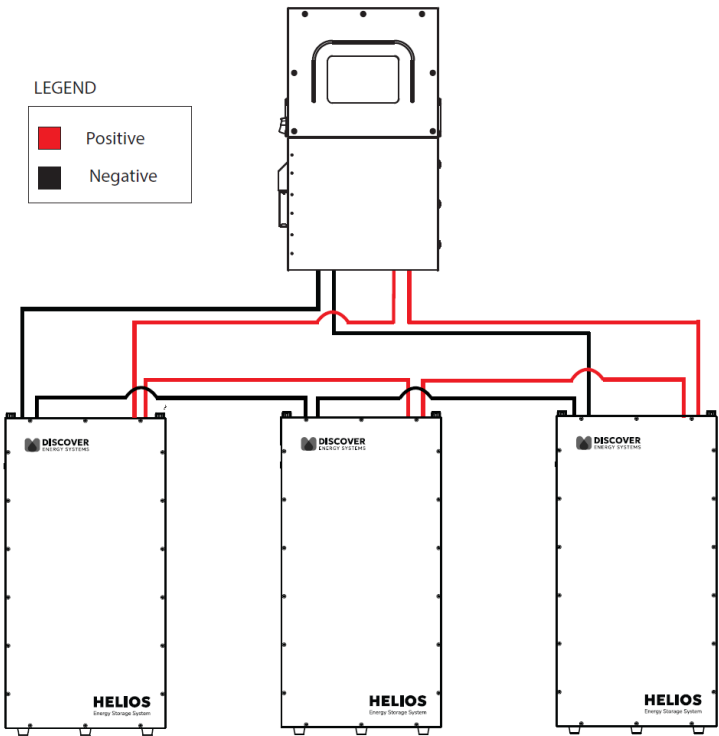


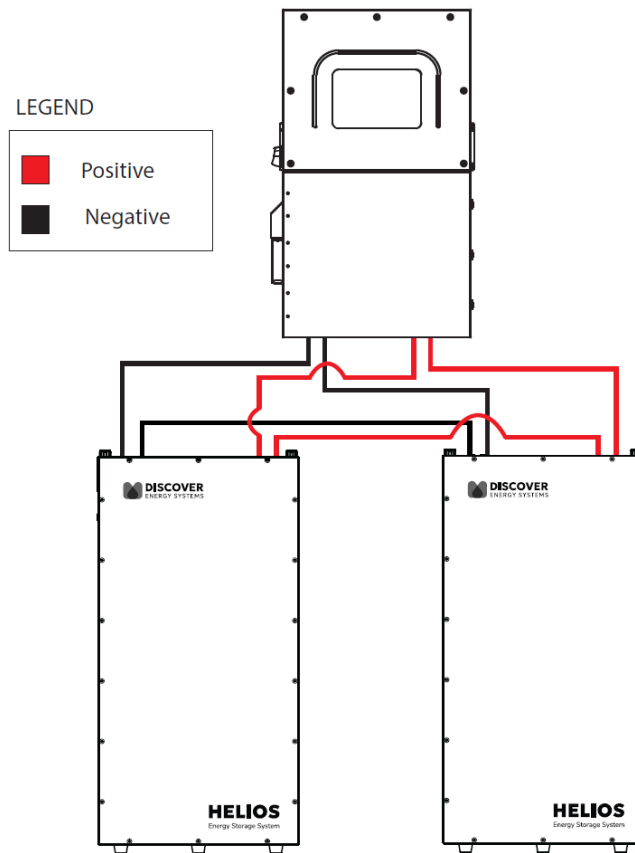
4. Connect the positive (+) battery cable in the HELIOS ESS Parallel Wire Kit (950-0068) to a receiving pin of the positive battery terminals on a battery at one end of the battery bank. Push the ends of cables against the receiving pins to secure the connection.



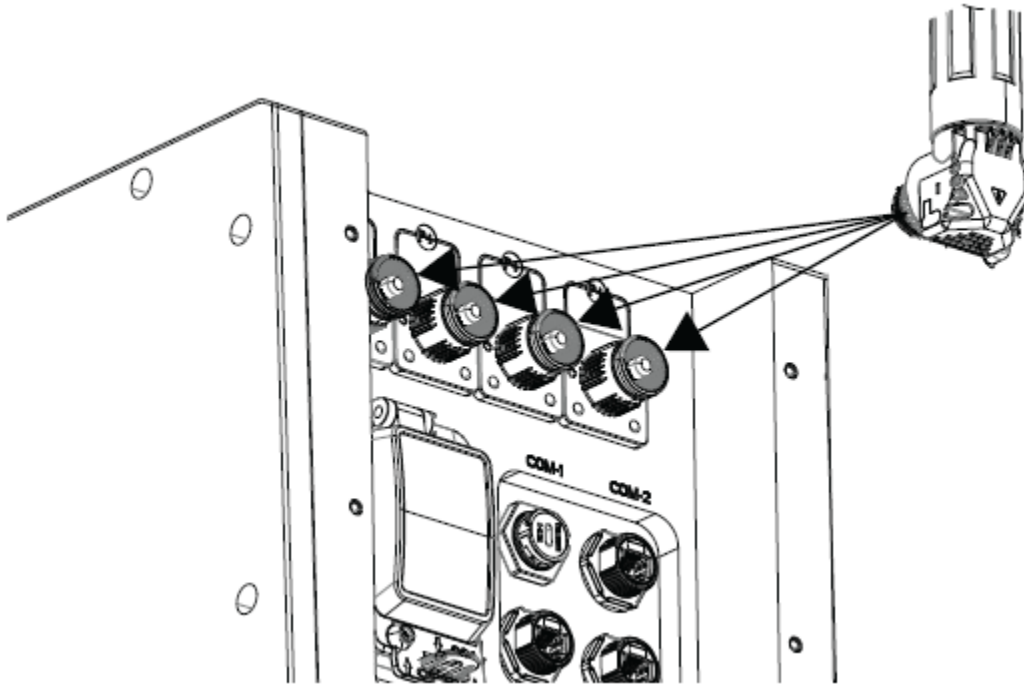
5. Connect the negative (-) battery cable to a receiving pin of the negative battery terminals on the battery at the other end of the battery bank. Push the ends of cables against the receiving pins to secure the connection.
6. Connect the positive (+) and negative (-) battery cables in the HELIOS ESS Parallel Wire Kit (950-0068) to the receiving pins of the positive and negative battery terminals on batteries as shown in the diagrams. Push the ends of cables against the receiving pins to secure the connection.

When Using TWO Sets of Battery to Inverter Cables





4. Connect one of the positive battery cables from the inverter to a battery on one end of the battery bank, and connect the other cable to the battery on the other end of the battery bank. Push the ends of cables against the receiving pins to secure the connection. Refer to the [“Terminal Connections and Hardware”](#) section.



5. Connect one of the negative battery cables from the inverter to a battery on one end of the battery bank, and connect the other cable to the battery on the other end of the battery bank. Push the ends of cables against the receiving pins to secure the connection.
 6. Connect the positive (+) and negative (-) battery cables in the HELIOS ESS Parallel Wire Kit (950-0068) to the receiving pins of the positive and negative battery terminals on batteries as shown in the diagrams. Push the ends of cables against the receiving pins to secure the connection.
-
7. Connect the CAN communication cables.
 8. Set all battery BMS to ON (ON/OFF key).
 9. Energize the system by setting all battery breakers to ON (close).
 10. Close the circuit disconnect if it is open.

NOTICE**TERMINAL BURNOUT**

Failure to securely lock the ends of battery cables to the receiving pins on the battery will increase resistance and lower voltage, leading to burnout of the terminals and voiding of the warranty.

Failure to follow these instructions may result in equipment damage.

NOTE

- Whenever replacing an old battery in a parallel string, use a battery of the same age, model, capacity, and voltage.
- Before installing the battery in parallel, charge each battery to 100% SOC. Breakers, fuses, or other disconnects may open if the batteries are not at the same SOC.

3.5 Paralleling More Than Three Batteries

To connect more than three batteries in parallel, use busbars sized so that they are capable of handling the maximum total charge and discharge current capacity of the loads and batteries.

The inverter connection is not identified as this can vary depending on the inverter type, number of inverters, and distance from the busbar.

NOTICE**BALANCING BATTERY CELLS**

All DC positive and DC negative cables from the busbar to each battery should be the same gauge and same length.

Failure to follow these instructions may result in equipment damage.

LEGEND

